

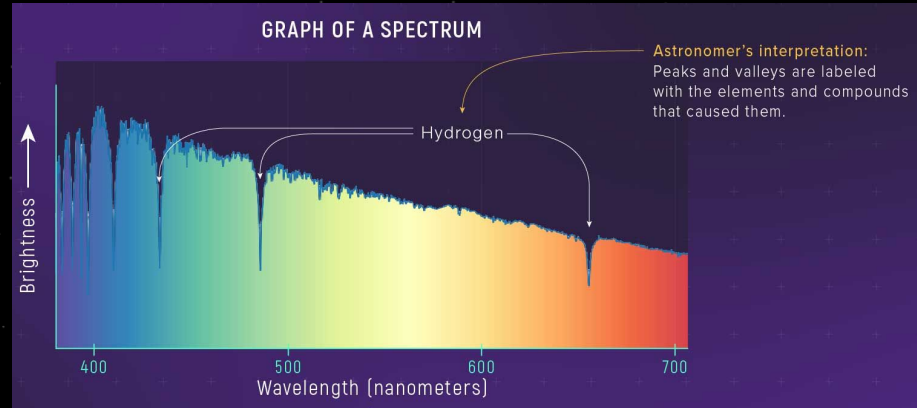
Deep Learning for Atmospheric Composition Estimation and Earth-like Exoplanet Identification

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Motivation

Our Goal: Build a model to predict the atmospheric abundances of exoplanets from spectra and explore K-Nearest Neighbors' ability to reduce a set of exoplanets to a smaller, more refined, optimistically habitable set

Current methodology using Bayesian Retrieval requires field expertise and high computation power. An ML pipeline allows for quicker analysis of new exoplanets as they're discovered and a reduced prospect set ensures efficient division of resources



Methods

- Developed a CNN to predict atmospheric abundances for :
 H_2O , CO_2 , N_2 , O_2 , CH_4 using simulated spectra as the feature space
 - ~18,000,000 trainable variables
 - Used MSE as our loss function
- Trained on a large synthetic dataset of terrestrial exoplanets
 - 273,000 example training set
 - 25,000 example validation set
 - 2,500 example test set
- Used K-NN to find the most Earth-like planets from the larger dataset

Results

- For our CNN, excluding N_2 our model produced an average MSE of 0.0683 and $R^2 = 0.9344$ for CO_2 , O_2 , H_2O , and CH_4
- We found K-NN to be a reliable way to prune the dataset, and using NASA estimates for how many expected habitable planets we should have in our dataset, the 75 closest planets to Earth shared remarkably similar features, suggesting a high potential for habitability

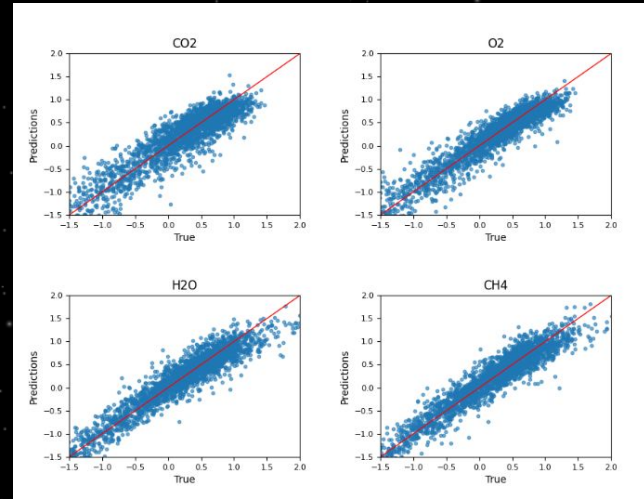


Table 1: Average feature values of Earth's 75 nearest neighbors

Feature	Mean Value	Earth Value
Planet Radius (km)	6289.51	6371
Surface Temp (K)	277.81	288
H_2O	0.0301	0.03
CO_2	0.0651	0.0004
O_2	0.1828	0.21
N_2	0.6748	0.78
CH_4	0.0141	0.00018

Takeaways

- A 1-D CNN is more than capable of digesting a spectra and outputting relative atmospheric abundances for molecules of interest
- K-NN as a method for reducing observations to a subset of interest is promising, and it capable of identifying Earth-like planets from an approximately uniform distribution

Future Work

- N_2 predictions could to be modeled better. High confidence nitrogen estimates are crucial in the determination of habitability
- The more data the better. We expect our CNN to get much better as a larger set of the 3 million planets is considered by those with access to stronger computing power